

The 65th ASH Annual Meeting Abstracts

ORAL ABSTRACTS

701.EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Microbiota-Specific T Cells Contribute to Graft-Versus-Host Disease after Allogeneic Stem Cell Transplantation

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Background:

Allogeneic T-cell expansion is the primary determinant of graft-versus-host disease (GVHD), and current dogma dictates that this is driven by histocompatibility antigen disparities between donor and recipient. This paradigm represents a fixed genetic system within which donor T cells interact with recipient-derived peptide-MHC complexes. However, the current model does not account for potential cognate interactions between donor T cells and microbe-derived antigen presented by the recipient. Prior studies in healthy patients have demonstrated the existence of circulating $\alpha\beta$ T cells targeting common bacterial species in the gastrointestinal (GI) tract. Whether microbe-specific T cells derived from the donor graft impact GVHD outcomes has not been studied. We tested the hypothesis that the recipient GI microbiota serve as a source of cognate antigen, contributing to clonotypic donor T-cell expansion and induction of GVHD.

Methods:

Functional studies utilized allogeneic stem cell transplant models in murine systems including both matched (Marilyn Tg to B6 male) and mismatched (B6 to B6D2F1) transplants as well as inducible MHCII knockout mice. Microbiota-specific transgenic T cells (MSTCs) targeting commensal flagellin (CBir1) and vendor-specific microbiome (segmented filamentous bacteria - SFB) in the context of I-A(b) were incorporated as part of the donor graft. Mechanistic studies utilized flow cytometry, cytokine stimulation and profiling, single-cell RNA sequencing, gene set enrichment analysis, NFAT-GFP T-hybridoma reporters, live-cell imaging (xCelligence RTCA eSight), and tissue analysis with RNA in-situ hybridization (ISH).

Results:

MSTCs lethally augmented GVHD when given as part of the donor graft in respectively colonized recipients. Taconic B6 male recipients (SFB-bearing) developed lethal GVHD when both donor Marilyn (Allo) Tg T cells and SFB T cells were introduced, but not when either transgenic T cell was introduced alone (**Figure A**, $p < 0.0001$). In contrast, Jackson B6 recipients (SFB-absent) receiving both Allo and SFB T cells did not develop lethal GVHD (**Figure A**, $p < 0.0001$), confirming functional relevance of microbe-specificity among the donor T cell pool. Lethal GVHD was also seen with addition of CBir1 T cells to the donor graft but not with addition of bulk T cells (log-rank $p < 0.0001$). Similar results were seen in a mismatched transplant system. To determine how MSTCs promoted lethality in the context of an allogeneic (allo) response, we performed scRNA sequencing of T cells harvested in the recipient spleen at day 6 post-transplant (prior to mortality onset). Post-transplant MSTCs harbored a naïve phenotype (S_{cell}^{high}/CD44^{low}/CD69^{low}) when given as the primary source of donor T cells but shifted to an early activated phenotype (S_{cell}^{int}/CD44^{low}/CD69^{high}/TCF7^{low}) when Allo T cells were concurrently introduced (ANOVA $q < 0.01$). MSTC activation was confirmed with augmented expression of IFN γ upon PMA/ionomycin stimulation observed in allo vs. syngeneic response (Wilcoxon rank-sum, $p < 0.01$). Introduction of Allo T cells increased the abundance of MSTCs in the mesenteric lymph nodes (day 5, 6, Wilcoxon, $p < 0.01$). Ablation of MHCII on the GI epithelial surface using Villin-CreERT2 I-Ab flox recipients abrogated microbe-specific T cell activation in co-culture with recipient GI intraepithelial cells harvested at day 6 post-transplant vs. non-ablated control (8, 12, 24-hr NFAT-GFP intensity, $p < 0.001$). RNA ISH probes targeting the CDR3 region of anti-microbial CBir1 T cells demonstrated an increase in MSTC infiltration at day 6 in the recipient GI lamina

propria during an allo response (**Figure B**) compared to a syngeneic response (# of MSTCs/terminal ileum cross section at day 6, Wilcoxon $p < 0.01$).

Conclusions:

We confirm that clonotypic T cell expansion following allogeneic transplantation is not only driven by donor-recipient genetics, as reflected by major and minor histocompatibility matching, but also by differences in microbiota composition that can serve as a source of antigenic stimulation, providing a feed-forward mechanism to propagate GVHD. Furthermore, in the context of an alloreactive T cell response that damages gut integrity, microbial antigens can be presented by GI epithelia, allowing the gut to be concurrently targeted by microbiota specific T cells whose function is additionally enhanced by the alloreactive response.

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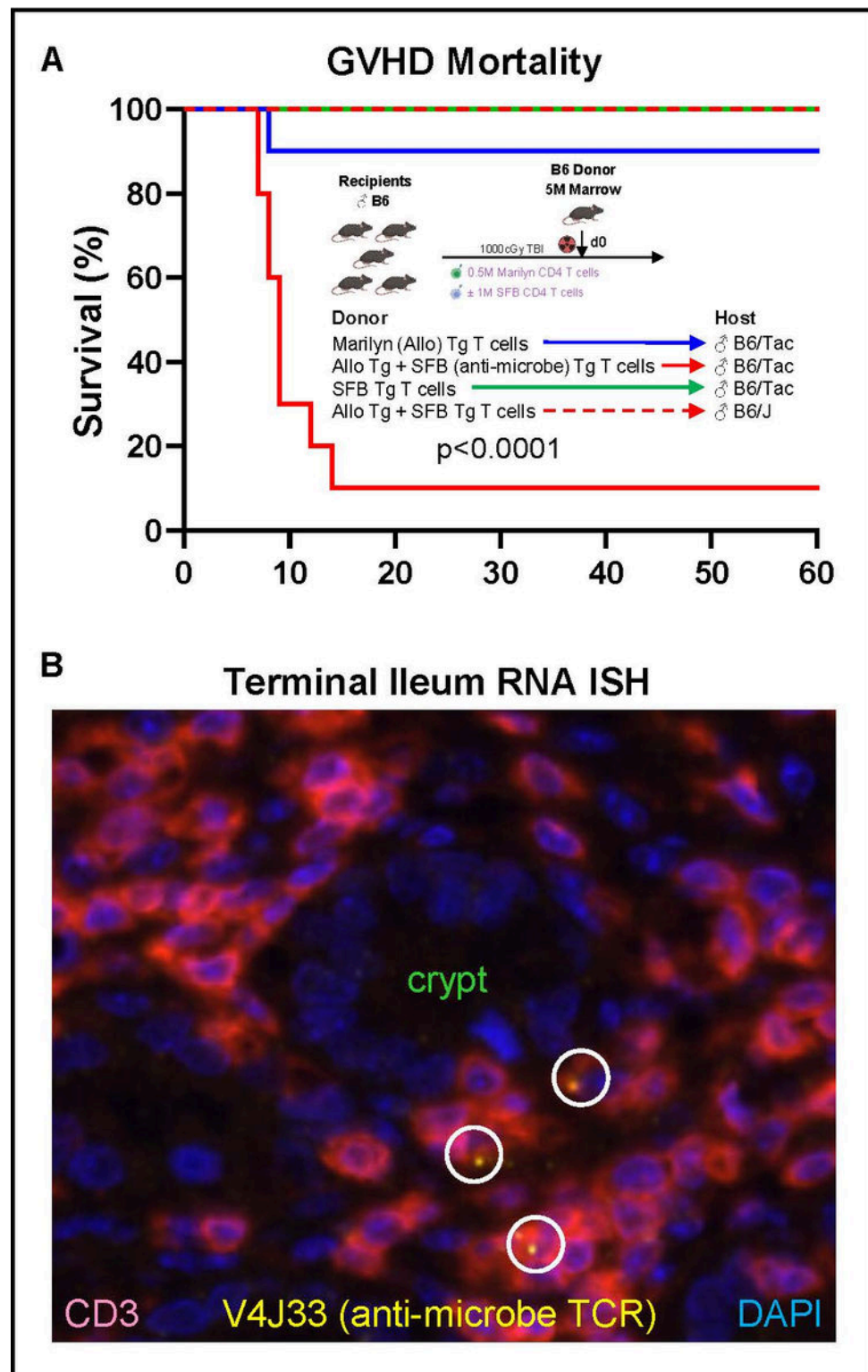


Figure 1

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